

Global Human Development Dashboard using UNDP HDI Data

Project Overview

This project analyzes global human development indicators using the Human Development Index (HDI) dataset published by the United Nations Development Programme (UNDP). The objective is to examine the relationship between human development and its key determinants, namely health, education, and income, and to develop an interactive analytical dashboard for country-level exploration and monitoring.

The project integrates data cleaning, exploratory data analysis, statistical modeling, machine learning, and interactive visualization techniques using Python and Google Colab.

Dataset

Source: United Nations Development Programme (UNDP) Human Development Report (HDR)

The original dataset contained HDI-related indicators for countries and territories across the world. Following preprocessing and cleaning, the final dataset consisted of:

- **204 countries/territories**
- **6 variables**

The variables used in the analysis are:

Variable	Description
Country	Country/Territory name
HDI	Human Development Index
Life_Expectancy	Life expectancy at birth
Expected_Years_Schooling	Expected years of schooling
Mean_Years_Schooling	Mean years of schooling

GNI_per_Capita

Gross National Income per
capita

Data Cleaning and Preprocessing

The original dataset contained section headers, mixed data types, and missing values that required preprocessing prior to analysis.

The following steps were performed:

- Removal of section headers such as *Very High Human Development* and *High Human Development*.
 - Selection of variables relevant to the analysis.
 - Conversion of object-type columns to numeric format.
 - Treatment of missing and invalid observations.
 - Creation of a cleaned dataset comprising 204 countries and 6 variables.
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Exploratory Data Analysis

Exploratory analysis was carried out to understand the distribution of HDI and its relationship with health, education, and income indicators.

Top 10 Countries by HDI

The analysis identified the following countries as the top performers in terms of Human Development Index:

Country	HDI
Iceland	0.972
Norway	0.970
Switzerland	0.970
Denmark	0.962
Germany	0.959
Sweden	0.959
Australia	0.958
Hong Kong (China SAR)	0.955

Netherlands	0.955
Belgium	0.951

These countries consistently exhibit high levels of health, education, and income, the three dimensions used in the calculation of HDI.

Correlation Analysis

A correlation heatmap was developed to examine the relationships among HDI and explanatory variables.

The analysis revealed:

- A strong positive association between HDI and Life Expectancy.
- Positive correlations between HDI and education indicators.
- A substantial positive relationship between HDI and GNI per Capita.

Overall, countries with stronger performance in health, education, and economic indicators tend to achieve higher levels of human development.

HDI Distribution

The distribution of HDI values was analyzed using histograms and summary statistics.

Key observations include:

- HDI values ranged from **0.388 to 0.972**.
 - The average HDI across countries was approximately **0.74**.
 - Most countries were concentrated in the medium to high HDI categories.
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Statistical Modeling

A Multiple Linear Regression model was developed to estimate HDI using the following explanatory variables:

- Life Expectancy
- Expected Years of Schooling
- Mean Years of Schooling
- GNI per Capita

The model achieved:

- **R-squared = 0.978**
- **Adjusted R-squared = 0.977**

The results indicate that approximately **97.8% of the variation in HDI** can be explained by the selected variables, demonstrating the strong influence of health, education, and income on human development outcomes.

Machine Learning Model

A Random Forest Classifier was developed to classify countries into:

- High HDI Countries ($HDI \geq 0.80$)
- Non-High HDI Countries ($HDI < 0.80$)

The model achieved:

- **Accuracy: 90%**
- **Weighted F1 Score: 0.90**

The results demonstrate that socioeconomic indicators can effectively distinguish between countries with high and non-high levels of human development.

Feature Importance

Feature importance analysis was performed to identify the variables contributing most significantly to HDI prediction.

The analysis indicated the following order of importance:

1. **GNI per Capita**
2. **Life Expectancy**
3. **Mean Years of Schooling**
4. **Expected Years of Schooling**

The findings suggest that economic prosperity is the strongest predictor of human development within the selected variables, followed by health and education indicators.

Interactive Dashboard

An interactive dashboard was developed in Google Colab using Plotly and ipywidgets.

The dashboard includes:

- Overview of HDI indicators
- Top and Bottom 10 countries by HDI
- Correlation heatmap
- HDI distribution
- HDI vs Life Expectancy scatter plot
- Country-level explorer using dropdown filters
- Feature importance visualization

The dashboard enables interactive exploration of human development indicators across countries and provides insights into the determinants of HDI.

Conclusion

This project demonstrates the application of data analytics, statistical modeling, machine learning, and interactive visualization techniques to analyze global development indicators.

The analysis highlights the strong relationship between human development and its core dimensions—health, education, and income. Both statistical and machine learning approaches indicate that economic prosperity, measured through GNI per Capita, is a major determinant of HDI, while health and education indicators also contribute significantly to development outcomes.

The project provides a data-driven perspective on global human development and illustrates how analytical techniques can support evidence-based decision-making and policy analysis in the development sector.